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The Editor-in-Chief

Classical and Quantum Gravity

IOP Publishing

Subject: Submission of Research Manuscript for Peer Review and Publication

Title of Manuscript: *Chrono-Energetics: An Emergent Approach to Spacetime Metric via Local Dissipation and Vacuum Stiffness*

Dear Editor-in-Chief,

I am writing to submit our original research manuscript entitled "**Chrono-Energetics: An Emergent Approach to Spacetime Metric via Local Dissipation and Vacuum Stiffness**" for consideration as a Research Article in *Classical and Quantum Gravity*.

As an engineer and alumnus of the French *Classes Préparatoires (Maths Sup / Maths Spé)*, my academic background has rigorously trained me in structural mechanics, continuous media, and mathematical formalization. Driven by a lifelong passion for foundational physics, I have dedicated my research to bridging the gap between thermodynamics and general relativity. This manuscript introduces a breakthrough concept, the **Theory of Chrono-Energetics**, which offers a fresh, mathematically solid perspective on the nature of time and gravity.

Our work departs from the traditional treatment of time as an invariant coordinate background. Instead, we propose that the local temporal flow (dt) is a macroscopic thermodynamic property emerging from the interaction between local information dissipation (dE_{diss}) and the intrinsic, high elastic tension of the quantum vacuum ($\mathcal{R}_v \approx 10^{43}$ N). Through precise dimensional analysis, we introduce the fundamental unit of emergent time, the **Chronom** (χ), demonstrating that time generation is intrinsically equivalent to a spatial metric extension ($1 \chi = 1 \text{ meter}$).

What makes this paper highly suitable for the readership of *Classical and Quantum Gravity* is its capacity to solve critical structural paradoxes in modern astrophysics using a single, unified formula:

1. **The Dark Matter Alternative:** We derive Milgrom's critical acceleration threshold (a_0) and the Baryonic Tully-Fisher relation from first principles.
2. **The Bullet Cluster Decoupling:** We formalize a geometric vacuum hysteresis effect using a finite relaxation time constant (τ_v).
3. **The Hubble Tension Resolution:** We demonstrate that a slow cosmic decay of global vacuum stiffness naturally resolves the H_0 discrepancy.

Furthermore, we propose three concrete, tabletop laboratory protocols to empirically test and measure the Chronom.

This manuscript is entirely original, has not been published elsewhere, and is not currently

under consideration by any other journal. Thank you very much for your time, consideration, and for assigning our work to expert peer review.

Sincerely,

Gregory Lombardo
*Independent Researcher in
Theoretical Physics
Graduate Engineer*